

Basic Game Theory

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Introduction

- Strategic Form Games
 - Individual Rationality
 - Mixed strategies
 - Common knowledge of Individual Rationality implies Iterative Strict Dominance
 - Nash equilibrium
 - Mixed Nash equilibria are weak
 - Knowledge conditions for Nash equilibrium
- Extensive Form Games
 - Subgame Perfection
 - Ultimatum game

Strategic Form Games

- The three elements of a strategic form game:
 - An index set of players $I = \{1, \dots, n\}$
 - A pure strategy space for each player S_i for all i in I .
 - A pure strategy assigns a feasible action to every information set controlled by the player.
 - Actions and information sets are represented in the extensive form.
 - An early result of analytical game theory is that the strategic and extensive forms should be nearly equivalent.
 - Payoff functions $u_i(s)$ that give player i 's von Neumann/Morgenstern utility for each strategy combination $s = (s_1, \dots, s_n)$ for all i in I .
- A strategic form game $G = \{I, S, U\}$, where

$$S = \times_{i \in I} S_i \quad U = \{u_i(s), \dots, u_n(s)\}$$

Parsimonious description

- The strategic form should strike you as an amazingly parsimonious description of a strategic situation.
- We have abstracted away from the context of the strategic situation.
- The strategy space hides enormous complexity even in parlor games like tic-tac-toe, chess, or go.

Payoffs

- Payoffs represent preferences over the lotteries defined on the consequences of the strategy combinations.
 - The consequences may be money, pride, reproductive fitness, pleasure, things, etc.
 - A key assumption is that these preferences can be expressed on a numerical utility scale
 - The scale must be at least ordinal.
 - When expected utility calculations are made the scale must be cardinal.

Common Knowledge of G

- Usually, the strategic form G is assumed to be common knowledge,
 - that is, all players know G ,
 - and know that their opponents know G ,
 - and know that their opponents know that they know,
 - and so on ad infinitum.
- This allows one to form rational models of how other players behave in situations represented by G .

Individual Rationality

- Individual rationality requires that player i choose a feasible strategy s_i^* that solves

$$\max_{s_i \in S_i} \sum_{s_{-i} \in S_{-i}} p(s_{-i}) u_i(s_i, s_{-i})$$

- where $p(s_{-i}) \in \Delta(S_{-i})$ is a subjective probability distribution over the actions of the other players,
- and $\Delta(S_{-i})$ denotes the set of probability distributions on S_{-i}
- that is, s_i^* is a *subjective best response* to $p(s_{-i})$.

Mixed Strategies

- Being unpredictable plays an important role in many strategic concepts.
- Unpredictability is formalized as a mixed strategy.
- A *mixed strategy*, denoted σ_i , is a probability distribution over pure strategies,
- that is, $\sigma_i \in \Delta(S_i)$, where $\sigma_i(s_i)$ is the probability assigned to pure strategy s_i .
- The *support* of a mixed strategy σ_i is the set of pure strategies to which σ_i assigns positive probability.
- It is conventional to use the same notation for the payoff to both pure and mixed strategies:

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s \in S} \left(\prod_{j=1}^n \sigma_j(s_j) \right) u_i(s)$$

Dominance

- Pure strategy s_i is *strictly dominated* for player i if there exists σ_i in $\Delta(S_i)$ such that $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})$ for all s_{-i} in S_{-i} .

Common Knowledge of Individual Rationality implies Iterated Strict Dominance



- b. Knowledge that player i is Savage rational rules out certain actions.
 - i. Specifically, those actions that are not a subjective best response to any $p(s_{-i}) \in \Delta(S_{-i})$.
 - ii. Such actions are strictly dominated actions.
 - iii. Let $S_i^0 = S_i$.
 - iv. $S_i^1 = \{s_i \in S_i^0 \mid s_i \text{ is a subjective best response to some } p(s_{-i}) \in \Delta(S_{-i}^0)\}$.
 - v. and $S_i^k = \{s_i \in S_i^{k-1} \mid s_i \text{ is a subjective best response to some } p(s_{-i}) \in \Delta(S_{-i}^{k-1})\}$.
- c. By finiteness, there must exist a K such that $S_i^k = S_i^K \neq \emptyset$ for all $k \geq K$.
- d. The sets $S_1^K, \dots, S_n^K$ are the sets of actions that the players can choose under the assumption of common knowledge of Savage rationality.
- e. This solution concept will be denoted *iterative strict dominance*.

Example One



	L	M	R
U	4, 3	5, 1	6, 2
M	2, 1	8, 4	3, 6
D	3, 0	9, 6	2, 8

(U,L) survives iterative strict dominance.

Example Two



	L	R
U	26, 46	10, 9
D	19, 11	35, 31

- Iterative strict dominance often fails to make a unique prediction.

Nash Equilibrium

- A *Nash Equilibrium* in a game is an assignment to each player of a strategy that is a best response when the other players use the strategies assigned to them.

- A pure strategy combination s^* is a *Nash equilibrium* if $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*)$ for all $s_i \in S_i$ and for all $i \in \mathbf{I}$.
- A mixed strategy combination σ^* is a *Nash equilibrium* if $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*)$ for all $s_i \in S_i$ and for all $i \in \mathbf{I}$.

Strict Nash Equilibrium

- A Nash equilibrium is *strict* if each player has a unique best response.
 - Strict equilibria are more compelling than weak equilibria, since one wonders why a player indifferent between several strategies conforms to the equilibrium assignment.
 - Strict equilibria are also robust to a number of small perturbations of the game.

Example Two Again

	L		R	
U	46	26	9	10
D	11	19	31	35

- (U,L) and (D,R) are strict Nash equilibria
- Thus, Nash equilibria predicts a subset of all possible outcomes.
- It is an area prediction theory.

“Consistent Predictions”

- Nash equilibria are “consistent predictions” of how the game will be played in the sense that if all players predict that a particular Nash equilibrium will occur no player has an incentive to play differently.
- A non-Nash strategy combination implies that at least one player will make a “mistake” either in his prediction of his opponents’ play or in his optimization of his payoff.

Knowledge for Pure Strategy Nash Equilibrium

- Theorem (Brandenberger, JEP 1992): Suppose that each player is Individually Rational and that the pure strategy choices of the players are mutual knowledge, then the choices constitute a pure strategy Nash equilibrium.

Prescriptive Theory

- Von Neumann and Morgenstern (1944) argued that any prescriptive theory that advises players how to play games must pick a mutually consistent strategy combination.
 - A theory recommending anything other than an equilibrium would be self-defeating in the sense that the player who believes that the others are following the theory will sometimes be motivated to deviate from it.
 - Note that this holds only if the theory recommends a unique strategy for each player.
 - If a game has several strict equilibria, what ought to be done depends on information not included in the strategic form of the game.

Strict Equilibria do not always exist in finite games.



Player 1:

U is a strict best response to R.

D is a strict best response to L.

	L	R	
U	0,0	0,-1	p
D	1,0	-1,3	$1-p$
	q	$1-q$	

Player 2:

L is a strict best response to U.

R is a strict best response to D.

- A. Hence, there are no dominated pure strategies.
- B. Moreover, there doesn't exist any pure strategy combination that is mutually consistent.
- C. Notice the best response cycle: U is a strict best response to R; L is a strict best response to U; D is a strict best response to L; R is a strict best response to D; U is a strict best response to R; ...

Nash Existence Theorem



Nash proved that allowing weak equilibria (mixed strategy equilibria) guarantees the existence of a mutually consistent way to play finite games.

- D. The strategy combination $\{\sigma, \tau\}$, where $\sigma(U) = \frac{1}{4}$, $\sigma(D) = \frac{3}{4}$, $\tau(L) = \frac{1}{2}$, $\tau(R) = \frac{1}{2}$ is mutually consistent.

$$\begin{aligned} Eu_1(p, \tau) &= 0 & \forall p \\ Eu_2(\sigma, q) &= 0 & \forall q \end{aligned}$$

- E. But it is a weak Nash equilibrium. The players are indifferent between their pure strategies and any mixture over the pure strategies.

Mixed Equilibria are weak.



Theorem: Any equilibrium point $\sigma^* = (\sigma_1^*, \dots, \sigma_i^*, \dots, \sigma_n^*)$ having one or more mixed strategy components σ_i^* is a weak equilibrium point.

Proof:

- [1] The proof is to show that a mixed strategy is never a unique best response. There must exist other strategies that are a best response to σ_{-i}^* and, hence, the equilibrium point is weak.
- [2] Without loss of generality, let $a_1^i, \dots, a_k^i$ denote the k pure strategies in the support of σ_i^* , that is, $\sigma_i(a_j^i) > 0$ and $\sum_{j=1}^k \sigma_i(a_j^i) = 1$.
- [3] Since σ_i^* is a best response, we have

$$u(\sigma_i^*, \sigma_{-i}^*) = \sum_{j=1}^k \sigma_i(a_j^i) u(a_j^i, \sigma_{-i}^*) = \max_{\sigma_i \in \Delta(S_i)} u(\sigma_i, \sigma_{-i}^*) = \pi_i$$

[4] Suppose that the component strategy a_i^j is not a best response, that is, suppose

$$u(a_i^j, \sigma_{-i}^*) < \pi_i$$

[5] then there must exist at least one action such that

$$u(a_i^j, \sigma_{-i}^*) > \pi_i$$

[6] But then σ_i^* is not a best response to σ_{-i}^* , which would contradict our assumption.

[7] Hence, if σ_i^* is a best response we must have $u(a_i^j, \sigma_{-i}^*) = \pi_i$ for all actions that are taken with positive probability, that is, $j = 1, \dots, k$.

[8] Hence, all actions that occur with positive probability are contained in the set of strategies that are best responses to σ_{-i}^* and so is any probability mixture over those actions.

[9] Since, there is not a unique best response the equilibrium point is weak. ■

Camerer (2003, p. 29) on Mixed-Strategy Equilibria

- "... the behavioral interpretation of mixing strategies is dubious. By definition, a player desires to mix only when she is indifferent among pure strategies, which means she does not (strictly) desire to mix with particular probabilities; she just doesn't care what she does.
- Furthermore, one player's equilibrium mixture probabilities depend *only* on the *other* player's payoffs, which is odd."
- Digression on Nash's Dissertation p. 21-23.

Refinements of Nash Equilibria

- Multiple Nash Equilibria frequently arise, the refinement literature attempts to rule out some Nash Equilibria as not being "fully" rational.
- Robust to perturbations of strategies and payoffs.
 - See, van Damme Stability and Perfection of Nash Equilibria
 - Strict equilibria are robust to these perturbations.
- Focus here on the problem of capturing sequential rationality in the solution concept.
 - Selten argued Nash Equilibria may not be sequentially rational.

An Extensive-Form Game is

- A configuration of nodes and branches running without any closed loops from a single (root) starting node to its end (terminal) nodes;
- An indication of which node belongs to each player;
- Probabilities that “nature” uses to choose branches at random nodes;
- Collections of nodes called information sets;
- And utility payoffs at each end node.

Information Sets

- An *information set* for a player is a collection of decision nodes satisfying:
 - The player has the move at every node in the information set
 - And when the play of the game reaches a node in the information set, the player with the move does not know which node in the information set has been reached.
- A *subgame* is the continuation game from a singleton node—that is, a node which has no other nodes in its information set—to the end nodes which follow from that node.

Subgame Perfection

- A *subgame perfect equilibrium* is
 - a Nash equilibrium of the original game
 - and its continuation strategies define a Nash equilibrium in every subgame.

The Ultimatum Game

- Let the ultimatum game G be defined by the following:
 - An index set of players: $N = \{1, 2\}$
 - Feasible strategies for the first mover player 1:
 - $s_1 \in S_1 = \{0, 1, \dots, 100\}$ in pennies
 - The second mover either accepts or rejects the proposal. Hence, the feasible strategies for the second mover are functions from the set of demands made by player 1 to either yes or no:

P2's strategy is a function

$$s_2 : S_1 \rightarrow \{Y, N\}$$

Preferences

- Assume that the money earned indexes player preferences over outcomes:

$$u_i(x) = x$$

- $\pi_1 = s_1$ if $s_2(s_1) = Y$ and 0 otherwise.
- $\pi_2 = 100 - s_1$ if $s_2(s_1) = Y$ and 0 otherwise.

Nash Equilibria

- All divisions are Nash Equilibria!
- $NE(G) = \{(x, 100-x) \mid x \in S_1\}$
- Supported by the following strategy for Player 2

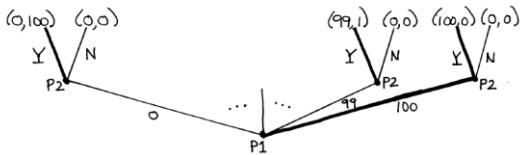
$$s_2(s_1)^* = \begin{cases} Y & \text{if } s_1 = X \\ N & \text{otherwise} \end{cases}$$

- Player 1's (weak) best response to $s_2(s_1)^*$ is $s_1^* = x$.
- Player 2's (weak) best response to s_1^* is Y .

Implausible Nash Equilibria

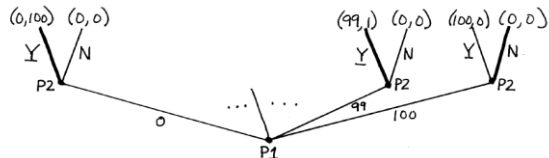
- One NE has player 2 receiving the entire dollar.
- $x = 0$
- Player 1 is indifferent between demanding 0 and demanding something more than 0, because he believes player 2 will reject $x > 0$.
- But would player 2 really reject a demand greater than zero?
- Nash Equilibria admit beliefs that aren't sequentially rational.

Extensive Form



- One Subgame Perfect Equilibrium is
- $s_2(s_i)^* = Y$ for all $s_i \in S_i$ and
- $s_1^* = 100$.

Another SPE



- Second SPE is
- $s_2(s_i)^* = Y$ for all $s_i < 100$, and $s_2(s_i)^* = N$ for $s_i = 100$.
- $s_1^* = 99$

Continuous Strategy Space

- For any finite grid there are 2 SPE, but the two SPE converge on the single SPE of the limiting infinite game $S_i = [0,100]$.
- Player 1 makes Player 2 just indifferent between accepting and rejecting.
- Player 1 gets the entire surplus.

References

- Fudenberg and Tirole *Game Theory*
- Brandenburger, "Knowledge and Equilibrium in Games" *Journal of Economic Perspectives* 6(4) Fall 1992.
